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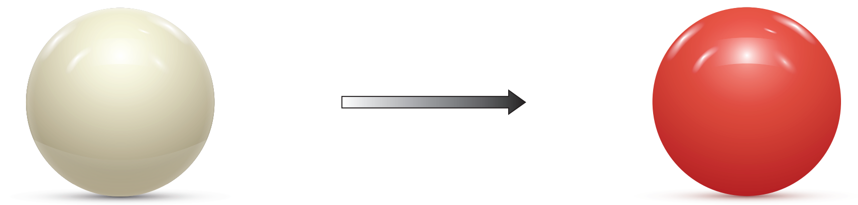
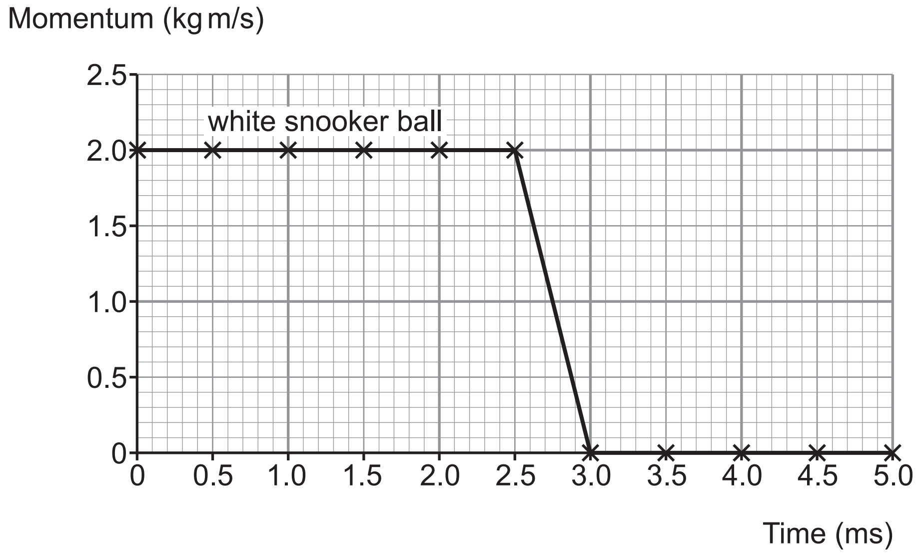
10. A white snooker ball, rolling to the right, collided with a **stationary** red snooker ball at a time of 2.5 ms.
Each snooker ball has a mass 0.16 kg.



The graph below shows the momentum of the **white snooker ball** before and after the collision.



(a) Use information from the graph to answer the following questions.

(i) State the initial momentum of the white snooker ball. [1]

initial momentum = ... kg m/s

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(ii) Use the equation:

$$\text{initial velocity} = \frac{\text{initial momentum}}{\text{mass}}$$

to calculate the initial velocity of the white snooker ball. [2]

initial velocity = ... m/s

(iii) The collision takes a time of 0.5 ms. State this time in seconds. [1]

time = ... s

(iv) Use the equation:

$$\text{resultant force} = \frac{\text{change in momentum}}{\text{time}}$$

to calculate the resultant force on the white snooker ball during the collision. [2]

resultant force = ... N

(b) (i) Underline the correct word or phrase from each of the brackets below which correctly completes the sentence. [2]

The law of conservation of momentum states that the total momentum before a collision is **(less than / equal to / greater than)** the total momentum after a collision provided **(no / small / large)** external forces act.

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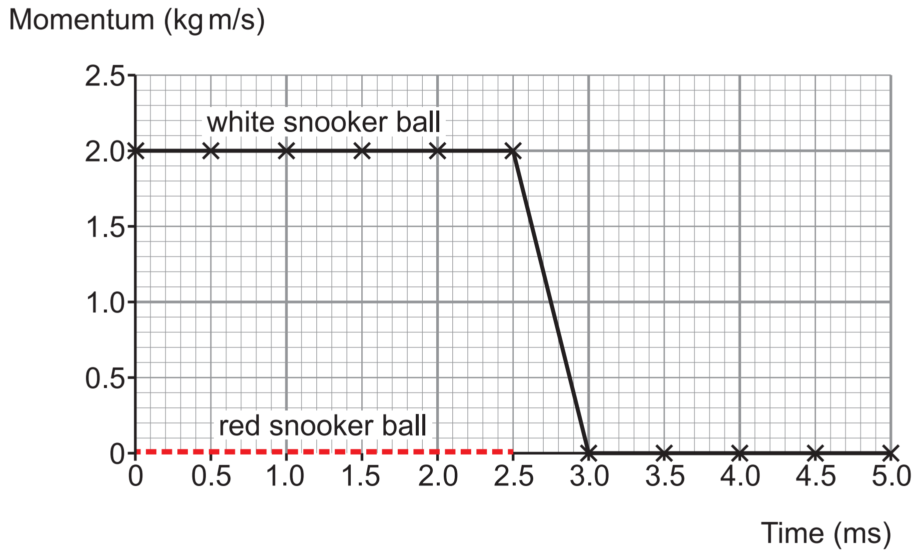
(ii) The momentum of the red snooker ball, between 0.0 and 2.5 ms, has been added to the original graph. It is shown as a red dotted line.
Complete the graph below to show the momentum of the red snooker ball from 2.5 ms to 5.0 ms. [2]



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